

A PROJECT REPORT ON
Crop Yield Prediction Using Machine Learning for Indian Agriculture

MACHINE LEARNING
BACHELOR OF TECHNOLOGY

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Abstract

This project develops a supervised machine-learning workflow for predicting crop yield from historical Indian agricultural data. The supplied CSV contains 19,689 records and 13 columns, including crop, year, season, state, area, production, rainfall, fertilizer, pesticide, temperature variables, and yield. Yield is the regression target. Production is excluded from the model inputs because it is directly related to yield and could introduce target leakage. After removing the single record with a missing target, 19,688 records remained. The input set contains 11 features: eight numerical and three categorical. Numerical missing values are imputed using the training-set median; categorical variables are one-hot encoded, and numerical variables are standardized within the preprocessing workflow. An 80:20 split with `random_state=42` produced 15,750 training records and 3,938 test records. Random Forest, XGBoost, and Support Vector Regression (SVR) were evaluated using MAE, MSE, RMSE, and R^2 . On the held-out test set, baseline XGBoost recorded MAE 10.3040, RMSE 125.3837, and R^2 0.9778; Random Forest recorded RMSE 152.4372 and R^2 0.9671, while SVR recorded RMSE 744.1601 and R^2 0.2163. XGBoost was therefore selected for the final demonstration among the evaluated baseline models. The results are specific to this dataset and random holdout split; further temporal and geographic validation is needed before real-world use.

Keywords: Crop Yield Prediction, Machine Learning, Random Forest, XGBoost, Support Vector Regression, Indian Agriculture.

1. Introduction

1.1 Purpose of the Project

Agriculture is important to food supply, livelihoods, and economic activity. Crop yield varies with crop type, season, cultivated area, rainfall, fertilizer and pesticide use, temperature, location, and other conditions. Estimating yield from historical observations can help analyze patterns in agricultural production and support planning. This project uses a tabular dataset of Indian agricultural records and trains regression models to estimate the numerical Yield value from available crop, geographic, temporal, and environmental predictors. The project is an educational predictive-modelling exercise; its outputs should be interpreted as estimates rather than guaranteed field-level outcomes.

The purpose of this project is to develop a machine learning-based crop yield prediction system that can analyze historical agricultural and environmental data and estimate the expected crop yield. The system is intended to support farmers, agricultural researchers, agricultural organizations, planners, and policymakers by providing data-driven information for better agricultural planning and resource management. Machine learning models can identify patterns and relationships among different input variables and use them to generate predictions for new data. Thus, the project aims to provide a systematic and efficient approach to crop yield prediction while supporting improved decision-making in agriculture.

1.2 Problem Statement

Crop yield prediction is treated as a supervised regression task. Given crop, year, season, state, area, rainfall, fertilizer, pesticide, and temperature variables, the system estimates the target column Yield. The dataset also contains Production, but this column is excluded from the predictors because it is directly related to yield and could leak information about the target. Model performance is assessed on a held-out test set using MAE, MSE, RMSE, and R^2 .

1.3 Existing Methods and Disadvantages

Traditional yield estimation can rely on historical averages, field observations, surveys, and domain expertise. Such approaches remain valuable, but they may not capture complex interactions among many predictors consistently. Machine-learning models can learn nonlinear patterns from historical records, although their results depend on dataset quality, feature availability, and validation design. Poor preprocessing, target leakage, or an unrepresentative test set can produce misleading performance estimates. This project therefore compares multiple regression algorithms and reports their measured results and limitations.

Machine learning approaches can address some of these limitations, but different algorithms have their own challenges. Simple regression models may not capture complex nonlinear relationships, while highly complex models can suffer from **overfitting** when the dataset is small or poorly prepared. Poor preprocessing, irrelevant features, missing values, and data leakage can also lead to misleadingly high performance during model evaluation. Some advanced models may require greater computational resources and may have limited interpretability. Therefore, appropriate preprocessing, feature selection, model validation, and comparison of multiple algorithms are required to develop a reliable crop yield prediction system.

1.4 Proposed System

The implemented workflow loads the supplied Crop Yield Data.csv file, inspects its schema and missing values, performs exploratory analysis, removes the record with a missing target, and excludes Production from the predictor set. The remaining numerical and categorical predictors are processed with a ColumnTransformer: numerical columns use median imputation and standardization, while categorical columns use most-frequent imputation and one-hot encoding. The data is split into training and test subsets before fitting the preprocessing transformer. Random Forest, XGBoost, and SVR are trained and evaluated on the same test split. GridSearchCV with three-fold cross-validation is used to tune Random Forest and XGBoost. The baseline XGBoost model is used in the final prediction demonstration because it has the lowest test MAE, MSE, and RMSE and the highest test R^2 among the three baseline models.

1.5 Research Objectives

1. **1. Prepare and inspect the supplied agricultural dataset, documenting its records, columns, target, and feature types.**
2. **2. Perform data preprocessing and exploratory data analysis, including missing-value checks, duplicate checks, encoding, scaling, and visual analysis.**
3. **3. Select predictors for crop-yield estimation and exclude Production to reduce the risk of target leakage.**
4. **4. Train and compare Random Forest, XGBoost, and Support Vector Regression models using a reproducible train-test split.**
5. **5. Evaluate the models using MAE, MSE, RMSE, and R^2 , and perform hyperparameter tuning for Random Forest and XGBoost.**
6. **6. Select a model based on held-out test results and demonstrate a prediction for a new input record.**

2. Literature Survey

The related-work summary below uses the four studies documented in the supplied literature-review file. Bibliographic details should be checked against the publisher records before submission.

No.	Title	Reference / Publication	Problem or Gap	Objective	Methods	Key Finding / Relevance	Future Research / Limitation
1	Improving wheat yield prediction through variable selection using Support Vector Regression, Random Forest, and Extreme Gradient Boosting	Moreno Sánchez et al.; Smart Agricultural Technology, 2025	Yield prediction depends on selecting informative crop and environmental variables.	Improve wheat-yield prediction through variable selection and model comparison.	Correlation analysis, PCA, SVR, Random Forest, XGBoost.	The supplied literature review reports strong XGBoost results for wheat; this is a crop-specific study.	Test the selected features and models on other crops and regions; verify full bibliographic details before submission.
2	Pre-harvest forecasting of rainfed wheat yield in Iran using multi-source remote sensing and machine learning	Bourbour et al.; Agricultural Water Management, 2025	Pre-harvest estimates must combine environmental information from multiple sources.	Forecast rainfed wheat yield in Iran.	Remote sensing and meteorological data; Random Forest, XGBoost, SVR; hyperparameter optimization.	The supplied review reports XGBoost as the strongest of the compared models in that setting.	The study is specific to rainfed wheat in Iran; verify full bibliographic details before submission.
3	Forecasting Crop Yield Anomalies on Panel Data via Spatially Demeaned Ensembles	Stracqualursi; Journal of Agricultural, Biological and Environmental Statistics, 2026	Spatial differences can complicate yield-anomaly forecasting.	Forecast crop-yield anomalies using panel data.	Spatial demeaning and an ensemble of Random Forest, XGBoost, and SVR.	Relevant to combining the same model families, but the target is yield anomalies rather than the direct Yield value used here.	Validate on different datasets and distinguish anomaly forecasting from direct yield prediction; verify full bibliographic details.
4	An end-to-end ML and XAI-based framework for crop yield prediction under agroclimatic variability	Laktionov et al.; Applied Soft Computing, 2026	Model performance and interpretability both matter under agroclimatic variability.	Build an end-to-end crop-yield prediction and explanation framework.	Preprocessing, feature selection, regression-model comparison, explainable AI.	The supplied review highlights the value of combining predictive modelling with interpretability.	This project does not implement XAI; verify full bibliographic details before submission.

3. Methodology

3.1 Proposed System Flow Diagram

Dataset loading (Crop Yield Data.csv)
↓
Schema, missing-value and duplicate inspection
↓
Remove the row with missing Yield
↓
Exclude Production from predictors
↓
80:20 train-test split (random_state=42)

↓
 Training-fitted numerical/categorical preprocessing
 ↓
 Train Random Forest, XGBoost and SVR
 ↓
 Tune Random Forest and XGBoost with 3-fold GridSearchCV
 ↓
 Evaluate using MAE, MSE, RMSE and R²
 ↓
 Compare baseline models
 ↓
 Select baseline XGBoost for the demonstration
 ↓
 Generate a sample crop-yield prediction

3.2 Dataset Description

The project uses the supplied file Crop Yield Data.csv. It contains 19,689 rows and 13 columns before cleaning. The target is Yield. The predictors used for modelling are Crop, Crop_Year, Season, State, Area, Annual_Rainfall, Fertilizer, Pesticide, Avg_Temperature, Max_Temperature, and Min_Temperature. Production is excluded from the model inputs to reduce target leakage. The dataset contains numerical and categorical variables; the notebook does not document a separate external download URL, so the source is recorded here as the dataset supplied with the project.

Dataset Property	Details
Dataset file	Crop Yield Data.csv (supplied with the project)
Initial dimensions	19,689 records × 13 columns
Rows after target cleaning	19,688
Problem type	Supervised regression
Target variable	Yield
Input features	11 (8 numerical, 3 categorical)
Excluded column	Production (potential target leakage)
Categorical features	Crop, Season, State
Numerical features	Crop_Year, Area, Annual_Rainfall, Fertilizer, Pesticide, Avg_Temperature, Max_Temperature, Min_Temperature
Missing values	9 initially; one missing-target row removed; remaining numerical predictor values imputed
Duplicate rows	0 detected
Processed feature count	99 after one-hot encoding

3.3 Experimental Setup

The experiments were developed in Python using Jupyter notebooks. Pandas and NumPy support data handling, Matplotlib and Seaborn support visual analysis, and scikit-learn provides preprocessing, model evaluation, and GridSearchCV. XGBoost provides the gradient-boosted regression model. The three evaluated algorithms are Random Forest Regressor, XGBoost Regressor, and Support Vector Regression.

Component	Specification
Language	Python
Environment	Jupyter Notebook
Data processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Preprocessing	scikit-learn Pipeline, ColumnTransformer, SimpleImputer, OneHotEncoder, StandardScaler
Models	Random Forest Regressor, XGBoost Regressor, SVR
Train/test split	80:20; random_state=42; 15,750 train / 3,938 test
Tuning	GridSearchCV, 3-fold CV; Random Forest and XGBoost
Metrics	MAE, MSE, RMSE, R ²

3.4 Data Preprocessing

The initial dataset has 19,689 rows and 13 columns. It contains nine missing cells: two each in Annual_Rainfall, Fertilizer, Pesticide, and Avg_Temperature, and one in Yield. The record with missing Yield is removed, leaving 19,688 rows. No duplicate rows were detected. The remaining predictor missing values are handled by median imputation in the numerical pipeline. The numerical features are standardized, and Crop, Season, and State are one-hot encoded. Production is excluded from X, and the split is performed before fitting the preprocessor. The fitted preprocessor transforms the 15,750 training rows and 3,938 test rows into 99 columns.

Preprocessing Steps:

1. Data inspection
2. Duplicate removal
3. Missing-value handling
4. Invalid-value and outlier analysis
5. Categorical encoding
6. Numerical scaling
7. Feature selection
8. Train/test splitting

9. Leakage prevention

3.5 Exploratory Data Analysis

Exploratory data analysis was used to inspect the dataset structure, data types, missing values, duplicate records, descriptive statistics, categorical-value counts, and the distribution of Yield. The dataset covers crop records from 1997 through 2020, with 55 crop categories, six seasons, and 30 states. Yield is strongly right-skewed: the median is 1.030 while the mean is 79.958, and the maximum is 21,105. This large spread indicates that a small number of very high-yield observations can influence squared-error metrics. The project also includes distribution and relationship visualizations for the agricultural variables.

Recommended EDA visualizations

- Crop-type distribution
- Crop-yield distribution
- Rainfall distribution
- Temperature distribution
- Cultivated-area distribution
- Rainfall vs. crop yield scatter plot
- Temperature vs. crop yield scatter plot
- Box plot of crop yield
- Correlation heatmap
- Feature importance plot after model training

3.6 Feature Engineering and Feature Selection

The target column is Yield. The model uses 11 predictors: Crop, Crop_Year, Season, State, Area, Annual_Rainfall, Fertilizer, Pesticide, Avg_Temperature, Max_Temperature, and Min_Temperature. Production is excluded because it is closely related to the target and could make evaluation unrealistically optimistic. The three categorical variables are one-hot encoded, and the eight numerical variables are median-imputed and standardized. After preprocessing, the input matrix contains 99 columns.

3.7 Model Building and Training

The task is regression because Yield is a continuous numerical target. Three models were trained: Random Forest Regressor, XGBoost Regressor, and Support Vector Regression with an RBF kernel. The baseline Random Forest used 100 trees and `random_state=42`. The baseline XGBoost model used 100 estimators, `learning_rate=0.1`, `max_depth=6`, `random_state=42`, and the squared-error objective. The baseline SVR used an RBF kernel, `C=100`, `gamma='scale'`, and `epsilon=0.1`. All models used the same processed train/test split.

3.8 Hyperparameter Tuning

GridSearchCV with three-fold cross-validation was used for Random Forest and XGBoost. The Random Forest search considered `n_estimators=[50, 100]`, `max_depth=[None, 10]`, and `min_samples_split=[2, 5]`. The selected

parameters were `n_estimators=100`, `max_depth=None`, and `min_samples_split=2`. The XGBoost search considered `n_estimators=[50, 100]`, `learning_rate=[0.05, 0.1]`, and `max_depth=[3, 6]`. The selected parameters were `n_estimators=100`, `learning_rate=0.05`, and `max_depth=6`. The tuned XGBoost test RMSE (134.0279) was worse than the baseline XGBoost RMSE (125.3837), so the baseline model was retained for the final demonstration. The preprocessing transformer was fitted on the training split before cross-validation; placing preprocessing inside the cross-validation pipeline would provide stricter fold-wise isolation.

3.9 Model Evaluation

Because Yield is continuous, the project uses regression metrics rather than classification metrics. MAE is the average absolute prediction error; MSE is the average squared error; RMSE is the square root of MSE and is expressed in the target's units; R^2 measures the proportion of target variation explained relative to a mean-baseline model. Lower MAE, MSE, and RMSE are preferable, while higher R^2 is preferable. The target's unit is not specified in the supplied notebook/data documentation, so no unit is asserted here.

- **MAE:** Measures the average absolute difference between actual and predicted yield.
- **MSE:** Measures the average squared prediction error.
- **RMSE:** Represents the square root of MSE and expresses error in the same unit as the target.
- **R^2 :** Indicates how much of the variation in crop yield is explained by the model.

Lower **MAE**, **MSE**, and **RMSE** indicate smaller prediction errors, while a higher **R^2** indicates better explanatory performance.

4. Results and Discussion

4.1 Preprocessing Results

The dataset initially contained 19,689 rows and 13 columns, with nine missing cells. One row with a missing Yield target was removed, leaving 19,688 rows; no duplicate records were found. The 80:20 split produced 15,750 training records and 3,938 test records. Median imputation handled missing numerical predictors, numerical features were standardized, and the three categorical features were one-hot encoded. The resulting processed matrices contained 99 features. The preprocessing transformer was fitted on training data and then applied to the test data.

4.2 Model Performance

Model	MAE	MSE	RMSE	R ²	Interpretation
Random Forest (baseline)	11.5883	23,237.0851	152.4372	0.9671	Strong baseline performance
XGBoost (baseline)	10.3040	15,721.0644	125.3837	0.9778	Best baseline result on this test split
SVR (baseline)	60.8075	553,774.2173	744.1601	0.2163	Substantially larger prediction errors
Random Forest (tuned)	11.5883	23,237.0851	152.4372	0.9671	Same reported test metrics as baseline
XGBoost (tuned)	11.3295	17,963.4885	134.0279	0.9746	Did not improve on baseline XGBoost test results

4.3 Visual Results and Discussion

The held-out test results are summarized in Table 3. XGBoost achieved the lowest MAE, MSE, and RMSE and the highest R² among the three baseline models. Random Forest also performed well, while SVR had substantially larger errors and a lower R². Actual-versus-predicted plots and residual plots were generated for all three models. The large maximum residuals and wide target range indicate that performance is not uniform across all observations; the overall metrics should therefore be interpreted alongside the plots and the target distribution. The reported results come from a random 80:20 split and do not establish performance on future years or unseen states.

5. Conclusion

This project implemented a crop-yield regression workflow using the supplied Indian agricultural dataset. The dataset was inspected, missing target data was removed, predictor missing values were handled through a preprocessing pipeline, and categorical variables were encoded. Production was excluded from the input features to reduce target leakage. Random Forest, XGBoost, and SVR were evaluated on the same held-out test set. Baseline XGBoost produced the strongest test results among these models (MAE=10.3040, MSE=15,721.0644, RMSE=125.3837, R²=0.9778) and was used for the final prediction demonstration. The sample input in the final notebook produced a predicted Yield of 2.6165. This is a model estimate for that input, not a guarantee of actual farm yield. The random-split results should be followed by temporal and geographic validation before practical deployment.

6. Future Work

1. Use larger and more diverse agricultural datasets covering additional crops, regions, seasons, and years.
2. Integrate additional environmental variables such as detailed soil properties, weather conditions, satellite imagery, and vegetation indices.
3. Evaluate additional machine-learning and ensemble algorithms to improve predictive performance.

4. Perform more extensive hyperparameter optimization using cross-validation and automated optimization techniques.
5. Investigate **Explainable AI (XAI)** techniques such as SHAP to make crop-yield predictions easier to understand.
6. Evaluate model robustness using external datasets and data from different geographical regions.
7. Investigate model drift when agricultural and climatic conditions change over time.
8. Develop a deployable web or cloud-based application that allows users to enter agricultural conditions and obtain crop-yield predictions.

7. References

1. Moreno Sánchez, J. C., Acosta Mesa, H. G., Trueba Espinosa, A., Ruiz Castilla, S., & García Lamont, F. (2025). Improving wheat yield prediction through variable selection using Support Vector Regression, Random Forest, and Extreme Gradient Boosting. *Smart Agricultural Technology*. Full DOI and issue details were not present in the supplied literature-review file.
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5. Dataset: Crop Yield Data.csv, supplied with this project. The notebook does record an external source URL.:<https://www.kaggle.com/datasets/apoorvagondane/crop-yield-data>
6. Optimizing Potato Yield Predictions in Uttar Pradesh, India: A Comparative Analysis of Machine Learning Models, *Scientific Reports*, 2025. The study compares Elastic Net, Random Forest, ANN, XGBoost and SVR.

Signature of the Guide

Course Instructor